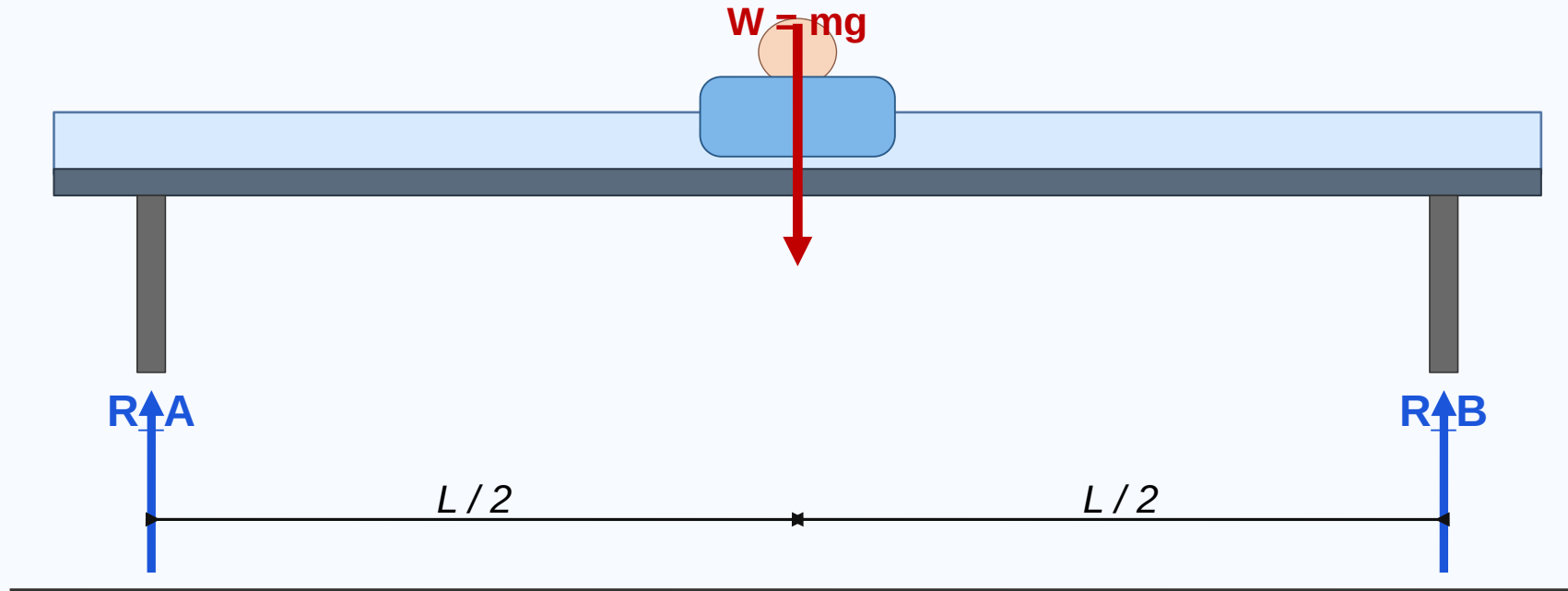


Statics for Medical Equipment

Lecture 4A - Forces, Balance, Supports, Moments

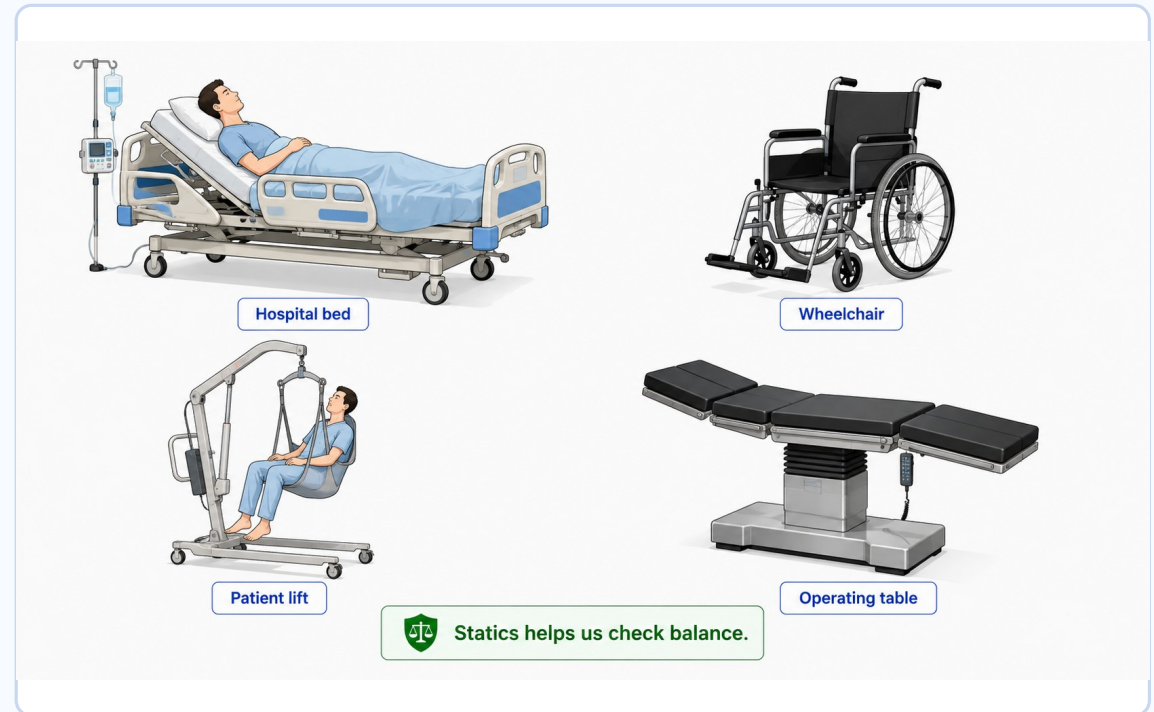


Statics = no movement, but forces are still working.

Why Do We Need Statics?

Hospital equipment must be safe.

- A bed must not break.
- A wheelchair must not tip over.
- A patient lift must not fall.
- An operating table must stay stable.



Statics helps us check balance.

From Materials to Mechanics

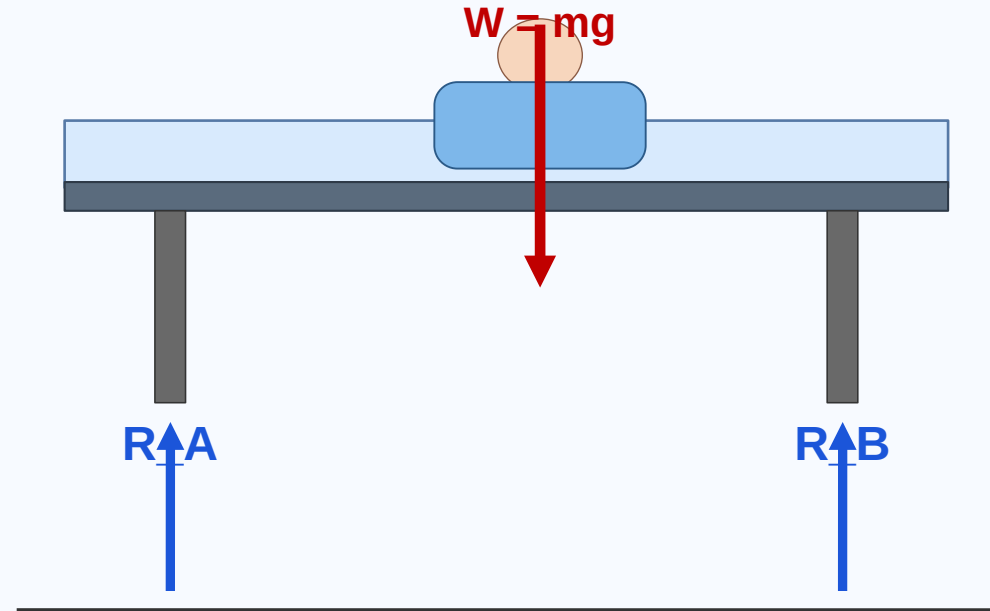
Last lesson: materials. Now: forces on materials.

Last lecture

What material? What property? What use?

Now

What force? What direction? What support? What balance?



A strong material is not enough. Shape and force also matter.

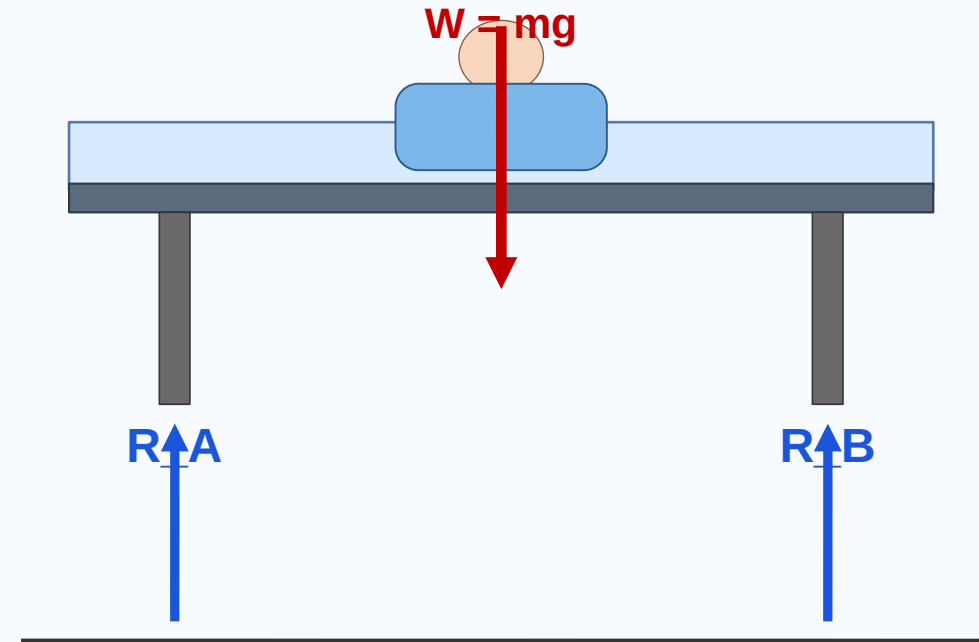
What Is Statics?

Statics studies objects at rest.

Statics = no movement

Forces are balanced

- The object is not moving.
- But forces still act on it.
- The total effect is zero.



Example

A patient bed is still, but the patient weight and support forces are acting.

Mass and Weight

Mass is not the same as weight.

Mass

Amount of matter
Unit: kg

Weight

Force from gravity
Unit: N

$$W = m g$$

- m = mass
- g = gravity
- In this class: $g = 10 \text{ m/s}^2$



kg is not a force. Newton (N) is a force unit.

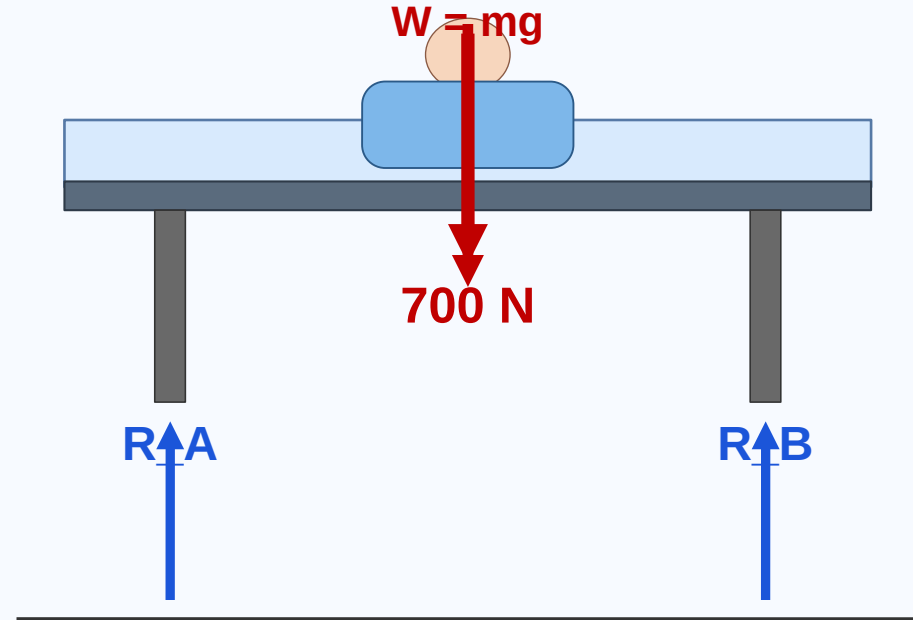
Example: Patient Weight

A patient has mass 70 kg.

$$m = 70 \text{ kg}$$

$$g = 10 \text{ m/s}^2$$

$$W = m g = 70 \times 10 = 700 \text{ N}$$

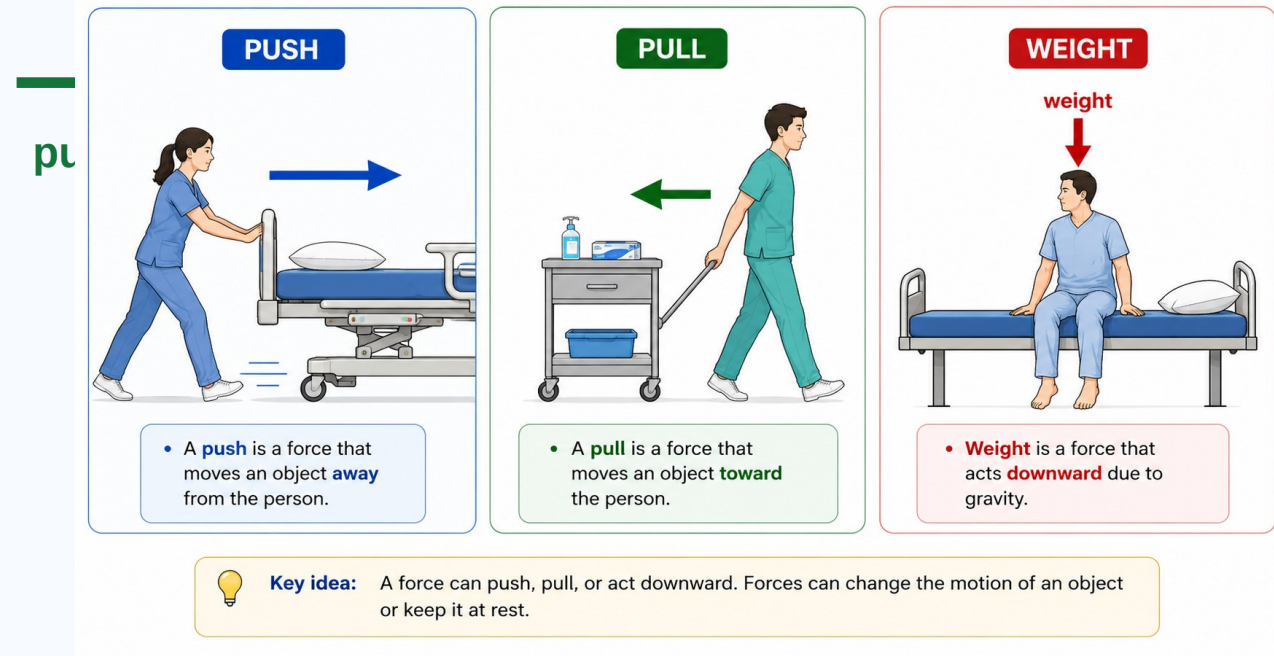


Weight acts downward.

What Is Force?

Force is a push or a pull.

- Force can push.
- Force can pull.
- Force can lift.
- Force can press.
- Force has direction.

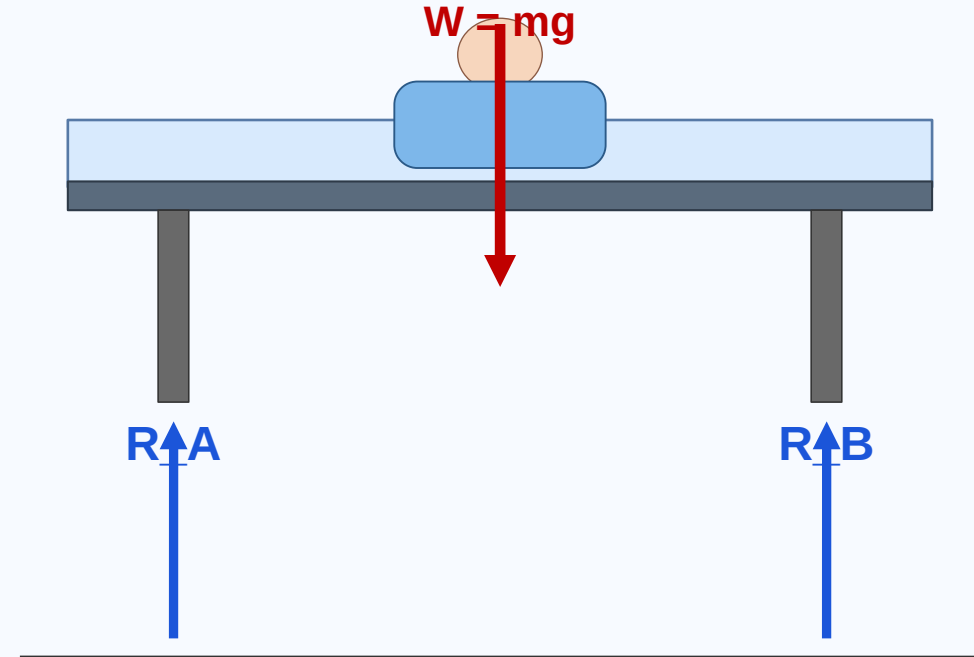


Force unit: N (newton)

Force Arrows

In mechanics, we draw forces with arrows.

- Arrow direction = force direction
- Arrow start = place of force
- Longer arrow = bigger force
- Label the force: W , R_A , R_B

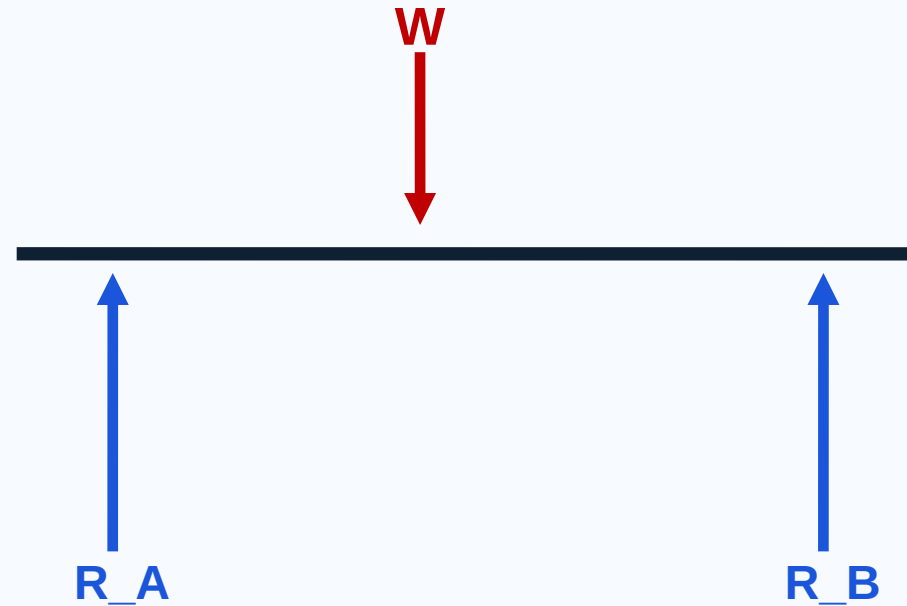


A force diagram is often more important than a long sentence.

Free-Body Diagram

Draw only the object and the forces on it.

- Remove extra details.
- Keep the important forces.
- Use arrows and labels.
- This helps solve the problem.



Idea

The free-body diagram is a simplified picture for calculation.

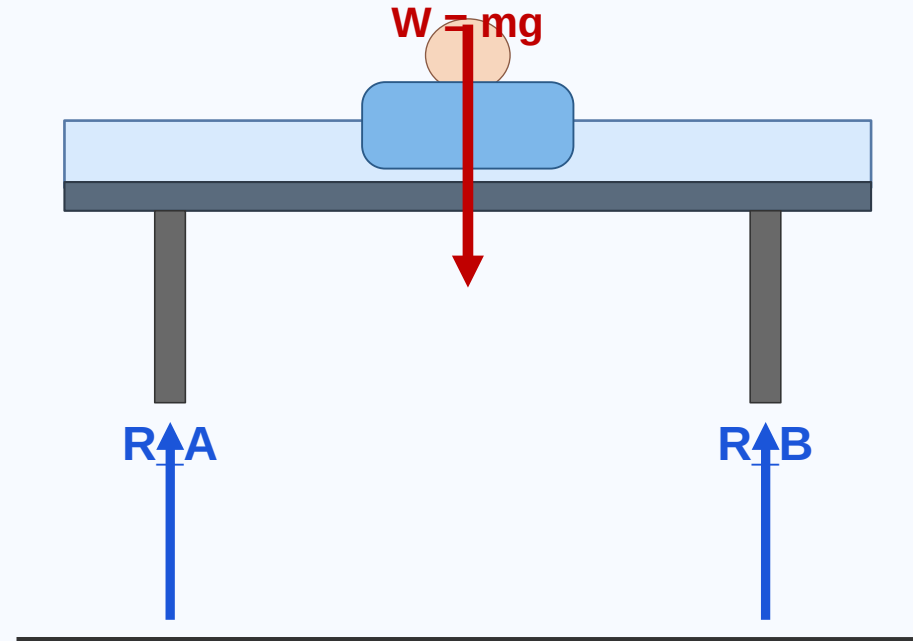
Equilibrium: No Vertical Movement

Upward forces and downward forces are balanced.

$$\Sigma F_y = 0$$

$$R_A + R_B - W = 0$$

$$R_A + R_B = W$$

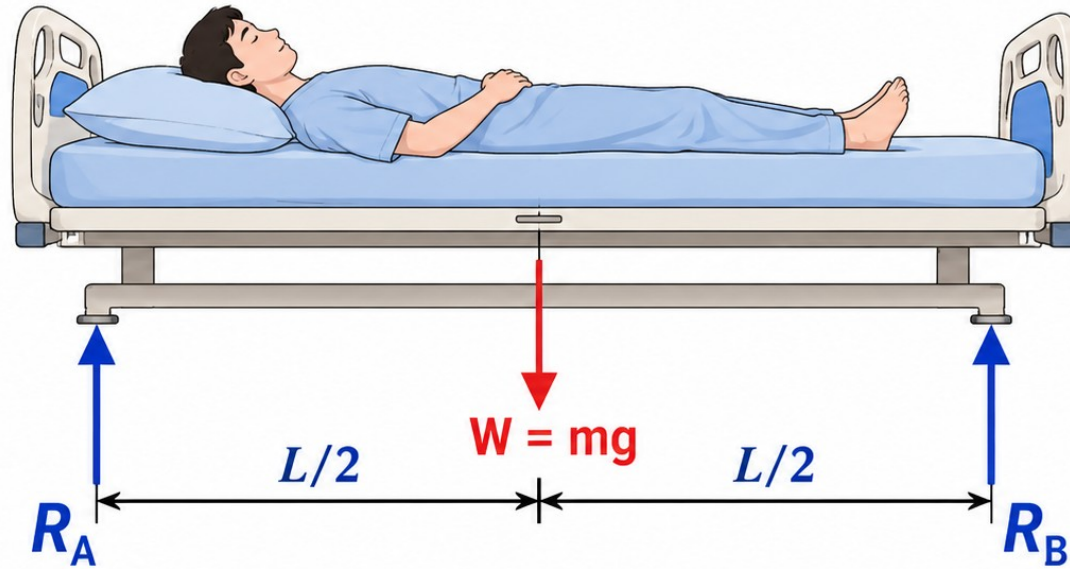


For vertical balance: total up = total down.

Patient in the Middle

A symmetric load gives equal support forces.

STATIC DIAGRAM: PATIENT IN THE MIDDLE




Patient in
the middle

 $R_A = R_B$



The bed is supported at two points (A and B).
The patient is in the middle. The reactions are equal.

N

Why Half and Half?

Use equilibrium and symmetry.

$$R_A + R_B = W$$

$$R_A = R_B$$

$$R_A = R_B = W / 2$$

Example

$$W = 700 \text{ N}$$

$$R_A = 350 \text{ N}$$

$$R_B = 350 \text{ N}$$

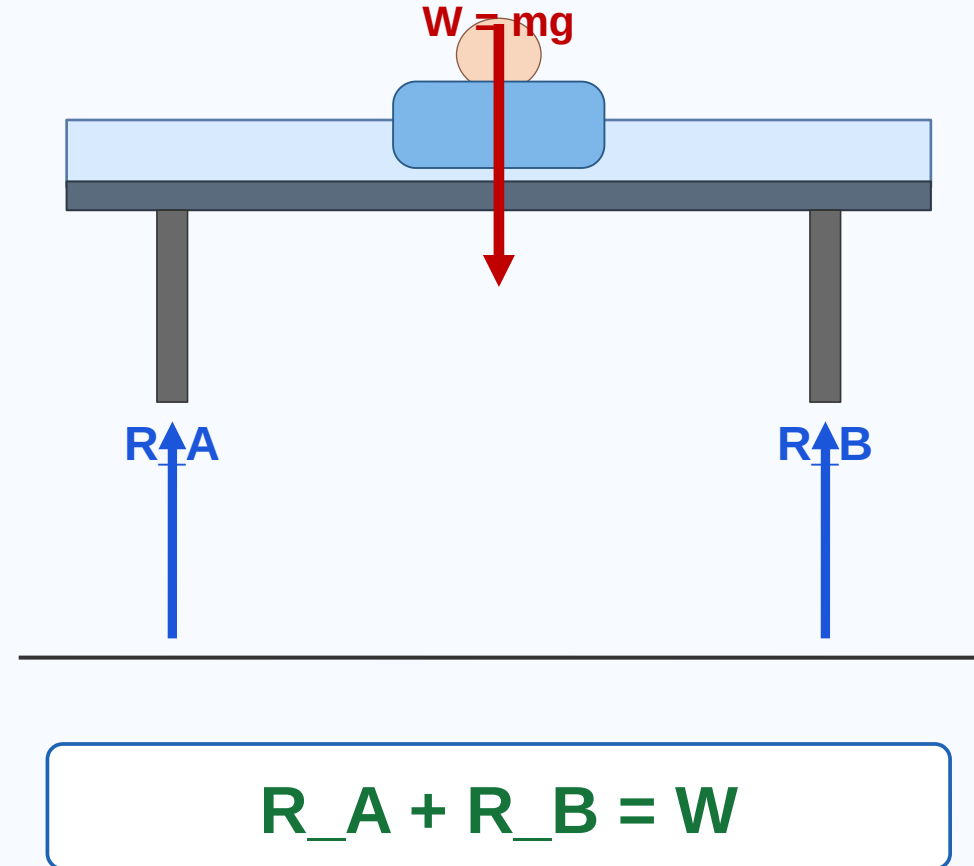


Only use half and half when the load is in the middle.

Support Reactions

A support reaction is a force from the support.

- The patient pushes down.
- The supports push up.
- The upward forces are reactions.
- We call them R_A and R_B .

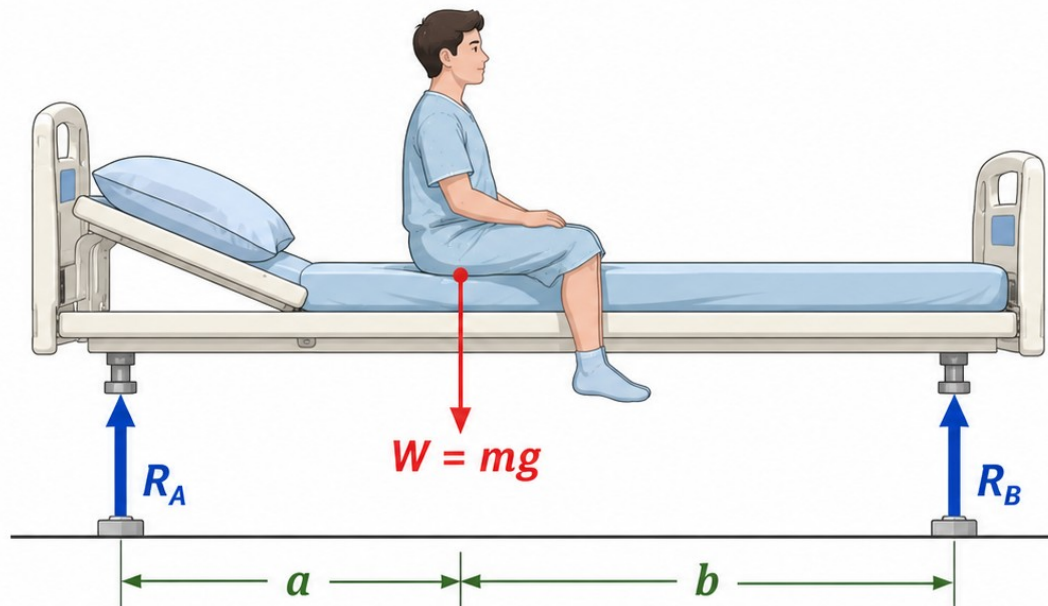


What If the Patient Is Not in the Middle?

Support forces are not equal.

Off-Center Load on a Simply Supported Bed

A patient sitting off-center creates unequal support reactions.



Patient is
not in the middle



Closer support
gets bigger force

Question

Which s



In static equilibrium: $R_A + R_B = W$ and $\sum M_A = 0$

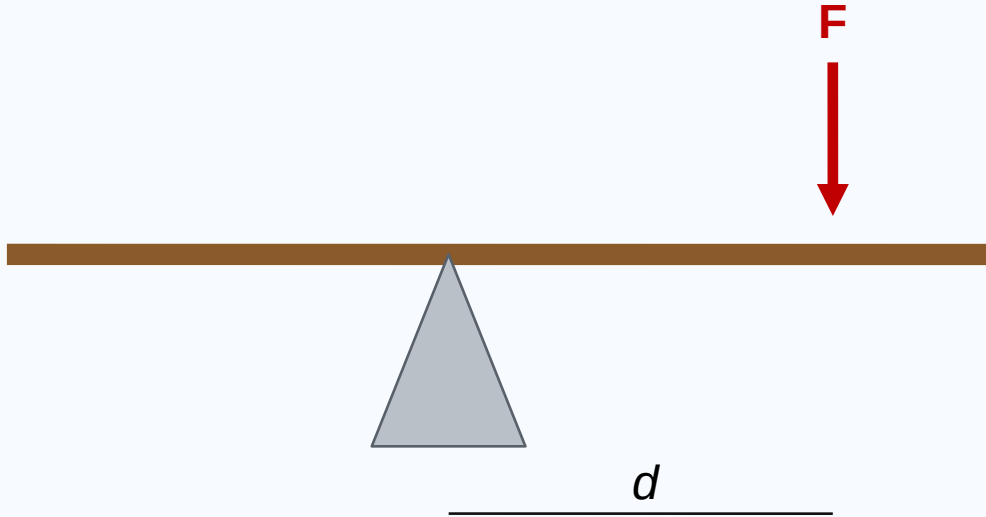
re.

New Idea: Moment of Force

Moment means turning effect.

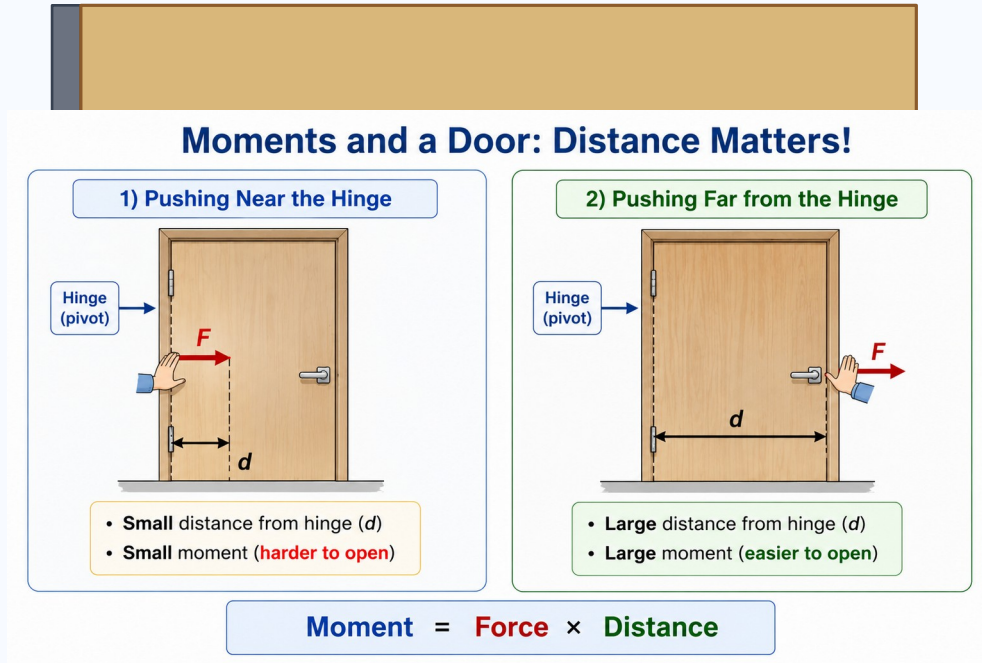
$$M = F \times d$$

- F = force
- d = distance from pivot
- Bigger force → bigger moment
- Bigger distance → bigger moment



Door Example

Same force, different distance, different moment.



hinge

Near hinge

Small distance
Small moment
Door is hard to open

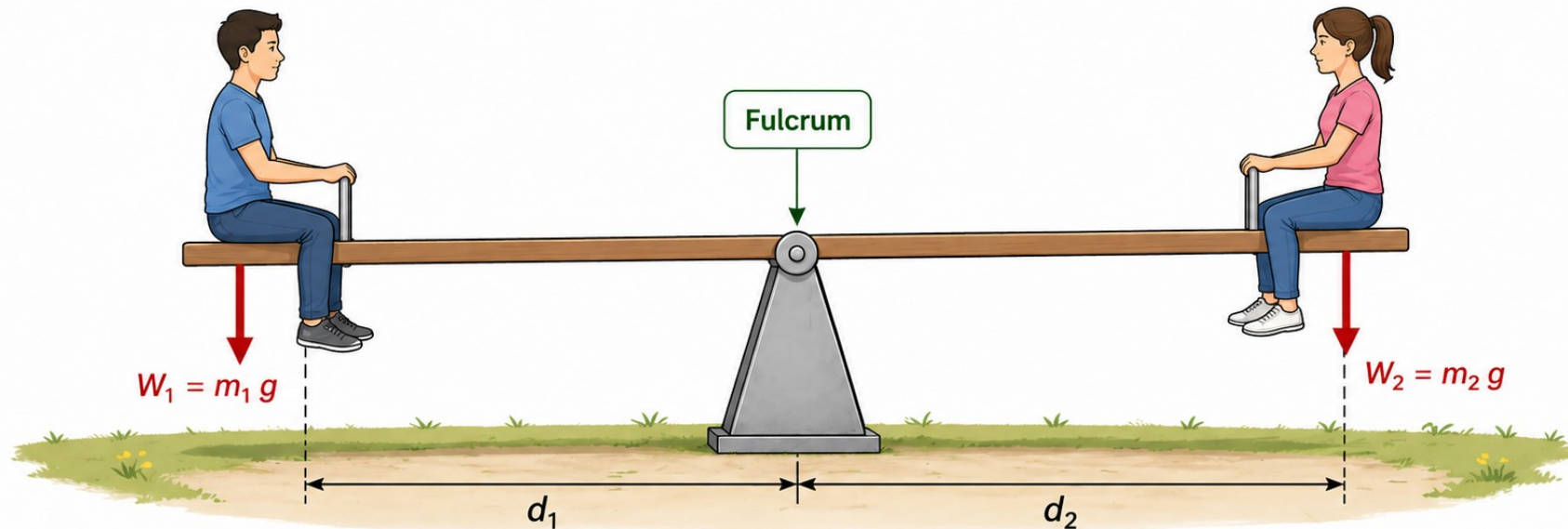
Far from hinge

Large distance
Large moment
Door is easy to open

$$\text{Moment} = \text{Force} \times \text{Distance}$$

Seesaw Rule

Balance means left moment equals right moment.



Rule of Moments (Balance Rule):

For equilibrium, the total clockwise moment equals the total counterclockwise moment.

$$W_1 \times d_1 = W_2 \times d_2$$

Note:



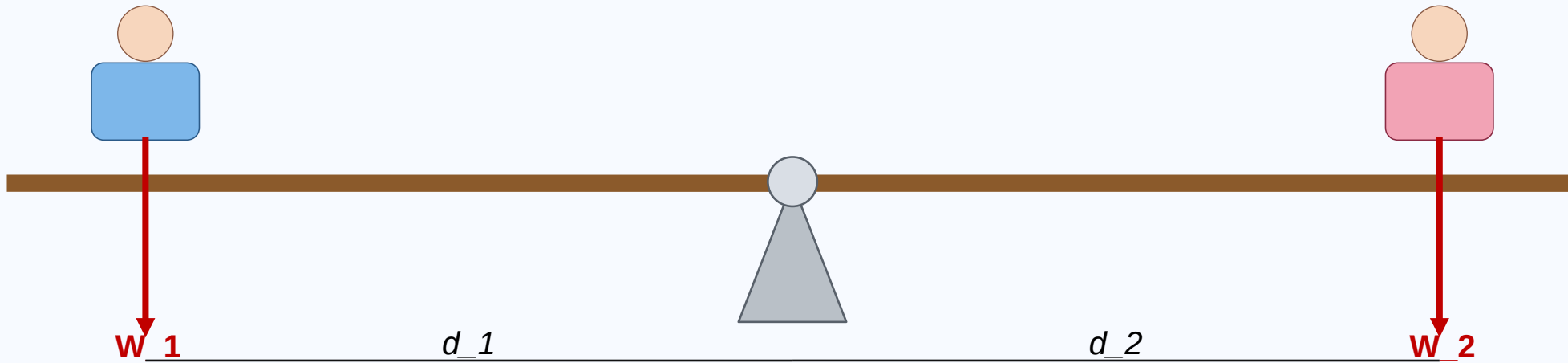
Heavy person → **closer**



Light person → **farther**

Heavy Person and Light Person

Balance is about force and distance together.



- Heavier person sits closer to the fulcrum.
- Lighter person sits farther away.
- Balance needs equal moments.

Large $W \times$ small d = Small $W \times$ large d

Two Conditions for Statics

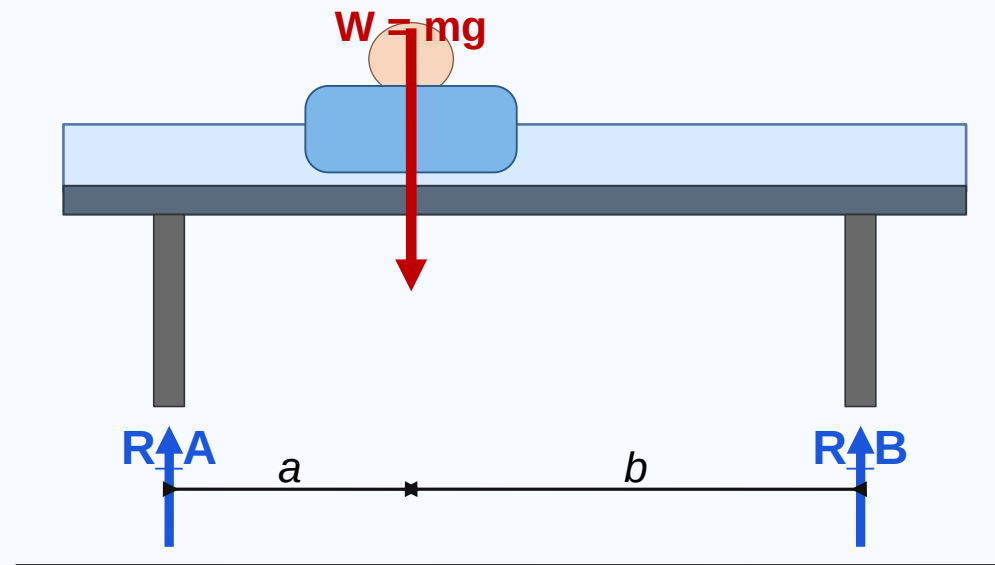
For full balance, we need two conditions.

$$1) \Sigma F_y = 0$$

No vertical movement

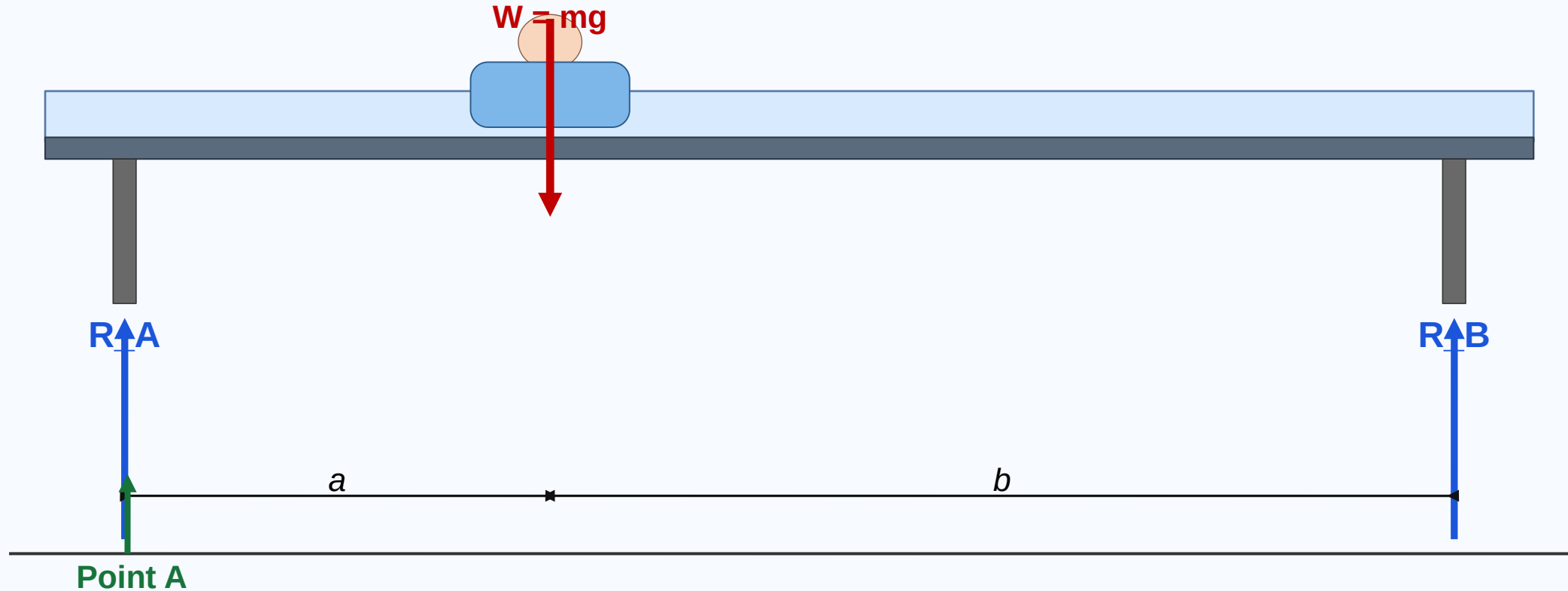
$$2) \Sigma M = 0$$

No rotation



Moment About Point A

Choose one support as the pivot.

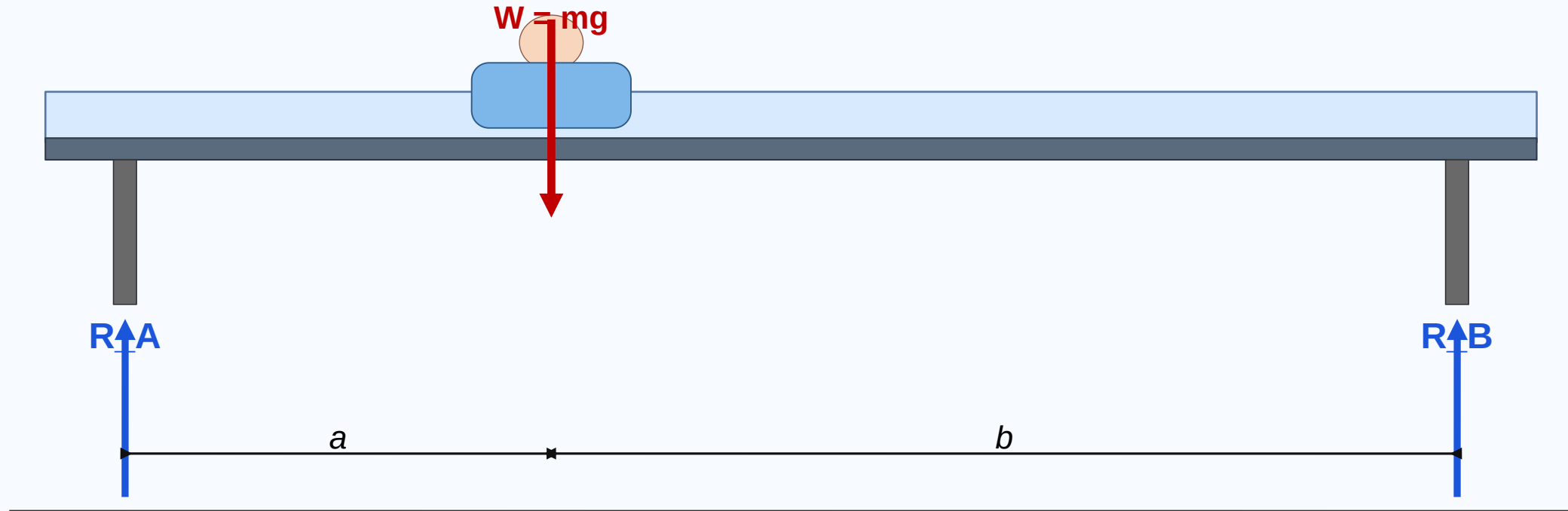


Moment from W: $W \times a$

Moment from R_B : $R_B \times (a + b)$

Support Force Formulas

For one point load between two supports.



$$R_A = W \times b / (a + b)$$

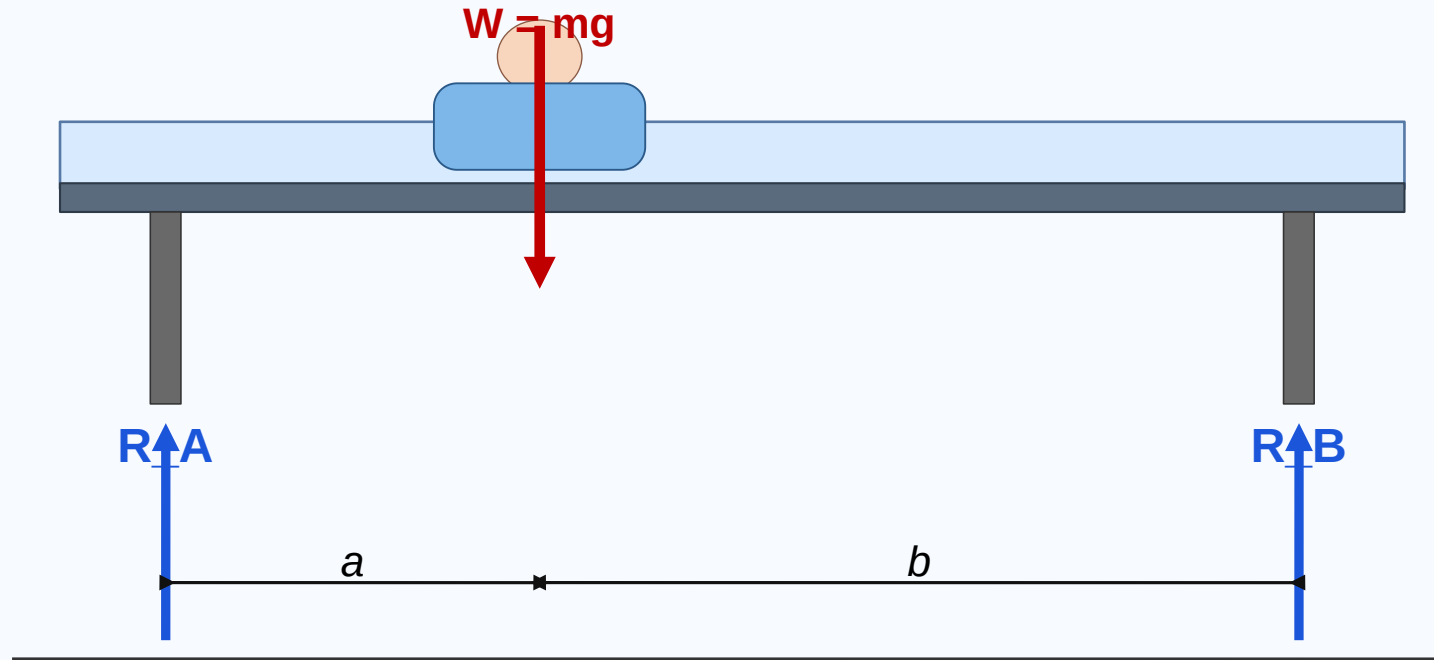
$$R_B = W \times a / (a + b)$$



Closer support gets bigger force.

Example Task: Patient Off-Center

Solve support reactions.



Given

$m = 70 \text{ kg}$
 $g = 10 \text{ m/s}^2$
 $a = 1 \text{ m}$
 $b = 2 \text{ m}$

Find

W
 R_A
 R_B



No solution yet. First draw the forces.

Solution Step 1: Weight and Total Force

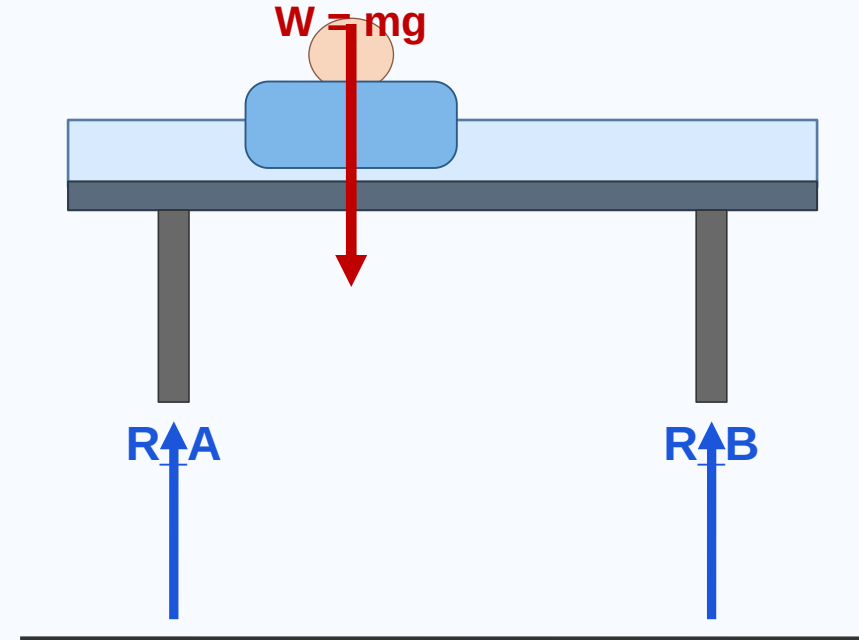
First calculate the weight.

$$W = m g = 70 \times 10 = 700 \text{ N}$$

$$R_A + R_B = 700 \text{ N}$$



This condition alone is not enough. We also need moment balance.



Solution Step 2: Moment Balance

Use the support force formulas.

$$R_A = W \times b / (a + b)$$

$$R_B = W \times a / (a + b)$$

$$R_A = 700 \times 2 / 3 = 467 \text{ N}$$

$$R_B = 700 \times 1 / 3 = 233 \text{ N}$$

$$\text{Check: } 467 + 233 = 700 \text{ N}$$



The patient is closer to A, so R_A is bigger.

General Rule for One Point Load

Use this when the load is between two supports.

$$R_A = W \times b / (a + b)$$

$$R_B = W \times a / (a + b)$$

- a = distance from left support to load
- b = distance from load to right support
- Closer support carries more
- Always check: $R_A + R_B = W$



Do not memorize only. Understand: moment balance creates these formulas.

Practice Task 1: Centered Patient

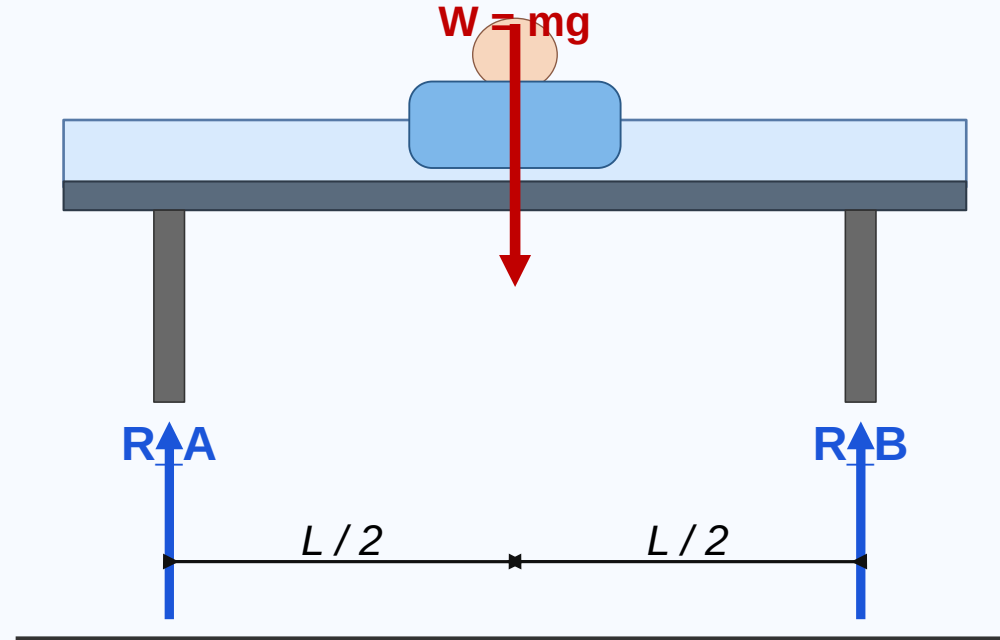
Try first. Then check.

Given

$m = 60 \text{ kg}$
 $g = 10 \text{ m/s}^2$
Patient is in the middle

Find

W
 R_A
 R_B



Answer: $W = 600 \text{ N}$; $R_A = 300 \text{ N}$; $R_B = 300 \text{ N}$.

Practice Task 2: Off-Center Patient

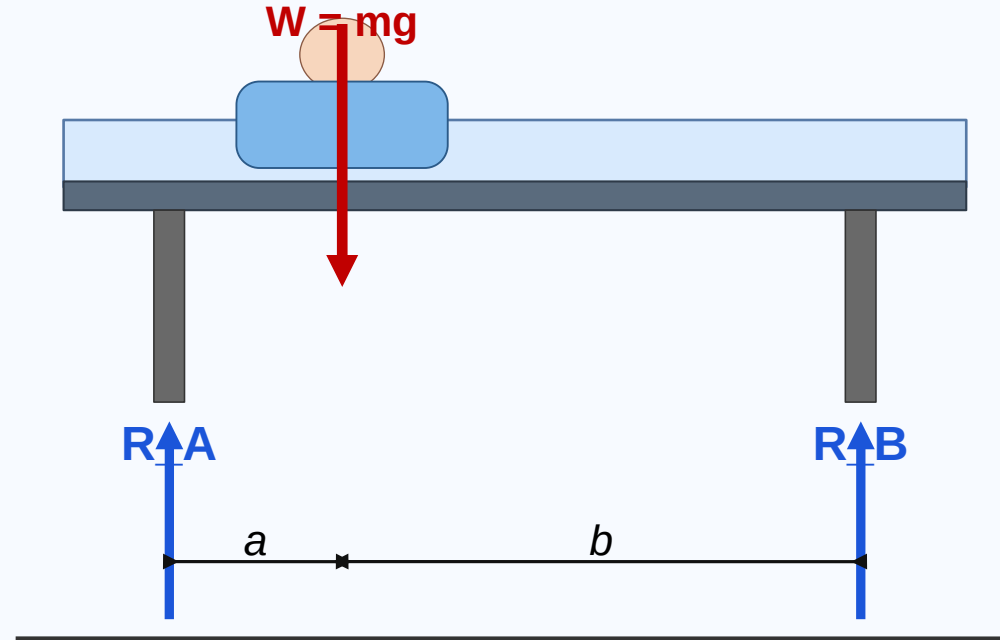
Use the formulas.

Given

$W = 800 \text{ N}$
 $a = 1 \text{ m}$
 $b = 3 \text{ m}$

Find

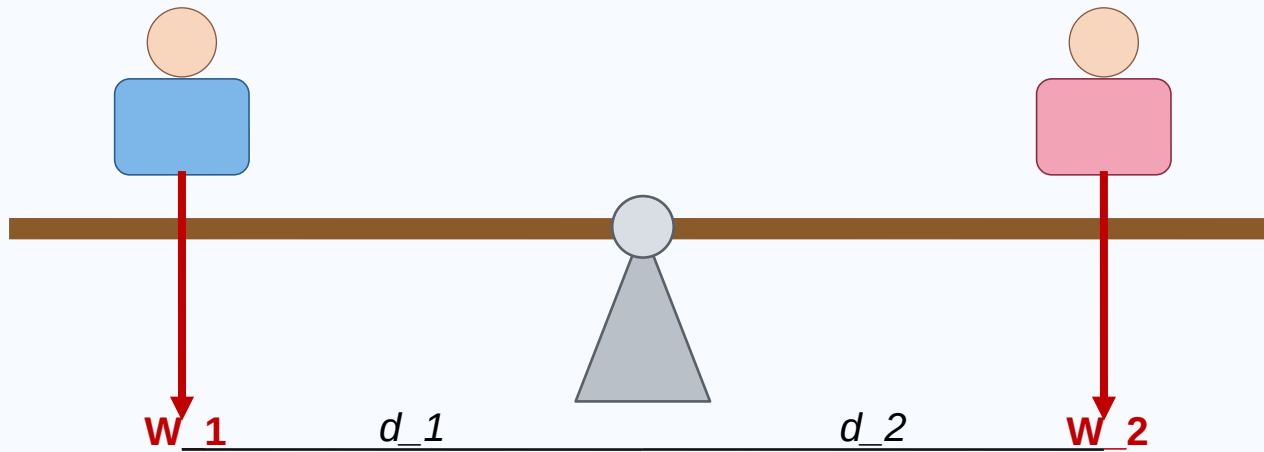
R_A
 R_B



Answer: $R_A = 600 \text{ N}$; $R_B = 200 \text{ N}$.

Practice Task 3: Seesaw

Find the missing distance.



Given

$$W_1 = 600 \text{ N}$$

$$d_1 = 2 \text{ m}$$

$$W_2 = 400 \text{ N}$$

Find

d_2

$$W_1 d_1 = W_2 d_2$$



Answer: $d_2 = 3 \text{ m}$.

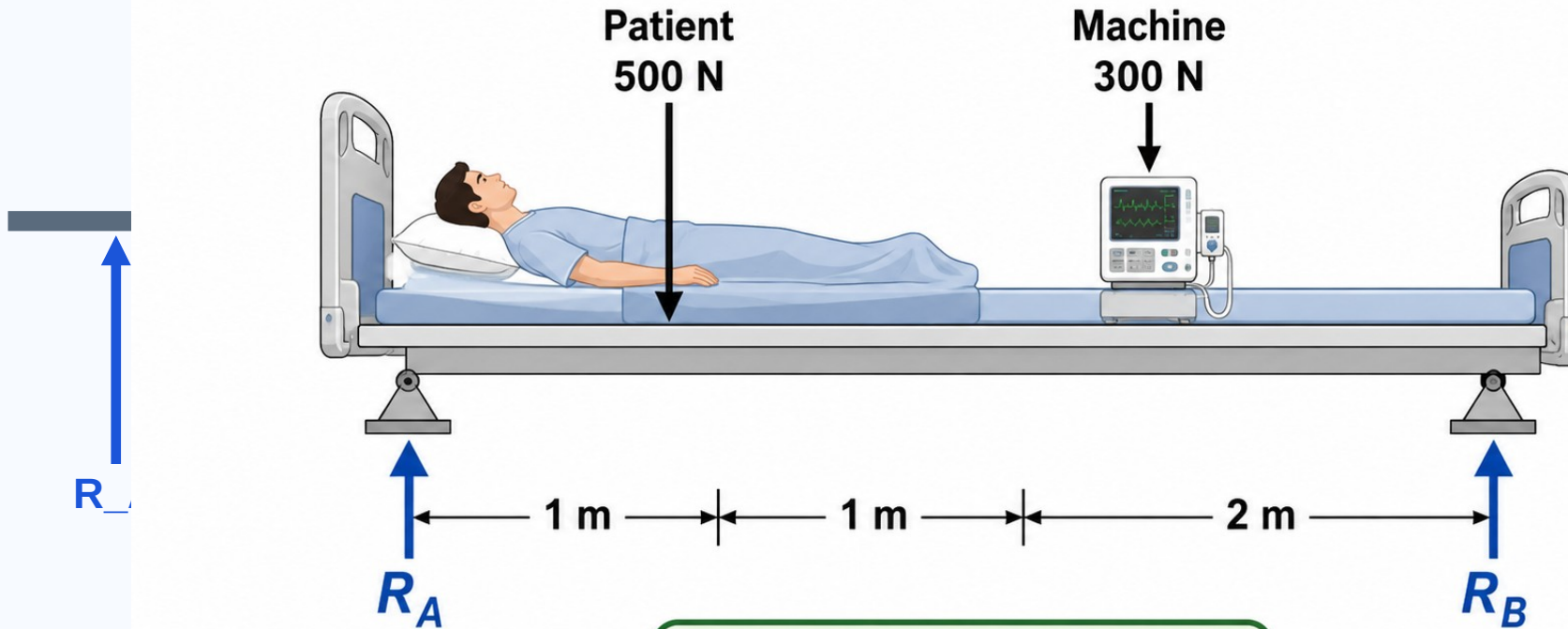
Practice Task 4: Bed + Equipment

Two downward forces.

Statics Task: Hospital Bed



$t = 1\text{ m}$
 $ne = 4\text{ m}$
 m



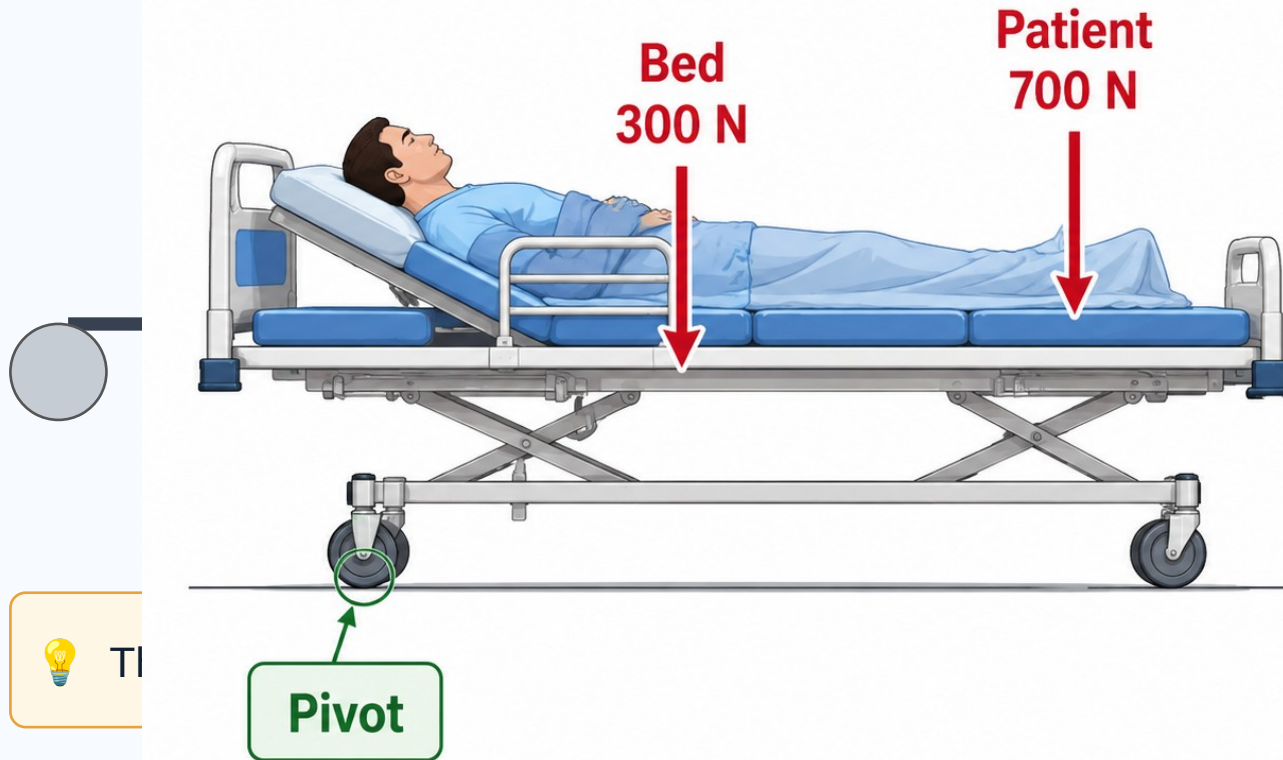
Find R_A and R_B

3 N; $R_A =$

Practice Task 5: Will It Tip?

Check moment around the wheel/support.

STATIC EQUILIBRIUM: WILL IT TIP?



Will it tip?



**Compare
moments
around the
pivot**

ound the
oment?

risk.

Common Mistakes

Avoid these errors.

- Using kg as force.
- Forgetting to multiply by g.
- Forgetting force direction.
- Thinking supports are always equal.
- Using only $\Sigma F = 0$ when rotation also matters.
- Forgetting to check the answer.

Check: $R_A + R_B = W$

Check: $\Sigma M = 0$

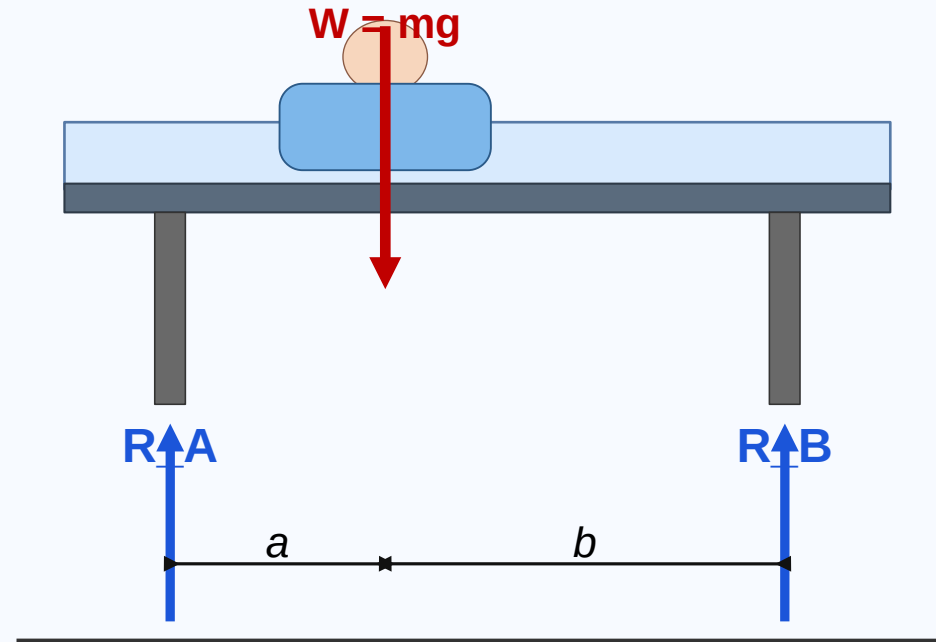


If the load is not in the middle, use moments.

Summary

Statics gives safety thinking.

- Weight: $W = mg$
- Equilibrium: $\Sigma F = 0$
- Moment: $M = F \times d$
- No rotation: $\Sigma M = 0$
- Support forces can be different
- Closer support carries more



Force → Balance → Moment → Safety